ANY H SHEAF TURNS AN ALMOST BLOWUP SQUARE INTO A CARTESIAN DIAGRAM OF SETS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0EVD (sources/stacks-flat/pinned-original-source.tex line 11216).

Allowed standard prerequisites (editorial).

- Surjective proper morphisms of finite presentation are h coverings, and the big and small h sites. Same chapter, Lemma labelled lemma-surjective-proper-finite-presentation-h (source line 10374) at proof lines 11231-11233 and Definition labelled definition-big-small-h (source line 10504) invoked in the statement. Supplies that the singleton family Z coproduct X' -> X is an h covering and which sites are meant.
- Almost blowup squares and their defining properties. Same chapter, Equation labelled equation-almost-blow-up-square (source line 10957), invoked in the statement. Supplies the square and the fact that b induces an isomorphism X' setminus E -> X setminus Z, which is used to build the second covering.

Source reference. Original definition 0EU1: sources/stacks-flat/pinned-original-source.tex, source lines 10504–10504. Same-chapter pointer into the author environment definition at source line 10504; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original equation 0EV6: sources/stacks-flat/pinned-original-source.tex, source lines 10957–10957. Same-chapter pointer into the author environment equation at source line 10957; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0ETW: sources/stacks-flat/pinned-original-source.tex, source lines 10374–10374. Same-chapter pointer into the author environment lemma

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at source line 10374; invoked at proof line 11233 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $\mathcal{F}$ be a sheaf on one of the sites $(Sch/S)_h$ constructed in Definition 0EU1. Then for any almost blow up square (0EV6) in the category $(Sch/S)_h$ the diagram*

$$\begin{array}{ccc} \mathcal{F}(E) & \longleftarrow & \mathcal{F}(X') \\ \uparrow & & \uparrow \\ \mathcal{F}(Z) & \longleftarrow & \mathcal{F}(X) \end{array}$$

is cartesian in the category of sets.

Proof. Since $Z \amalg X' \to X$ is a surjective proper morphism of finite presentation we see that $\{Z \amalg X' \to X\}$ is an h covering (Lemma 0ETW). We have

$$(Z \amalg X') \times_X (Z \amalg X') = Z \amalg E \amalg E \amalg X' \times_X X'$$

Since $\mathcal{F}$ is a Zariski sheaf we see that $\mathcal{F}$ sends disjoint unions to products. Thus the sheaf condition for the covering $\{Z \amalg X' \to X\}$ says that $\mathcal{F}(X) \to \mathcal{F}(Z) \times \mathcal{F}(X')$ is injective with image the set of pairs (t, s') such that (a) $t|_E = s'|_E$ and (b) s' is in the equalizer of the two maps $\mathcal{F}(X') \to \mathcal{F}(X' \times_X X')$. Next, observe that the obvious morphism

$$E \times_Z E \amalg X' \longrightarrow X' \times_X X'$$

is a surjective proper morphism of finite presentation as b induces an isomorphism $X' \setminus E \to X \setminus Z$. We conclude that $\mathcal{F}(X' \times_X X') \to \mathcal{F}(E \times_Z E) \times \mathcal{F}(X')$ is injective. It follows that (a) $\Rightarrow$ (b) which means that the lemma is true. $\square$